

Description

Students create a project called “Guess The Number” by implementing Python programming concepts they have learned. Students will create this project by using random module, functions, loops, user’s input, and conditional statements.

Objectives

1. Implementing all the python concepts students have gotten so far into a proper project.
2. Understanding the proper flow of creating a good project.
3. Advancing students experience in programming.
4. Reviewing codes and refactoring codes to be more efficient.

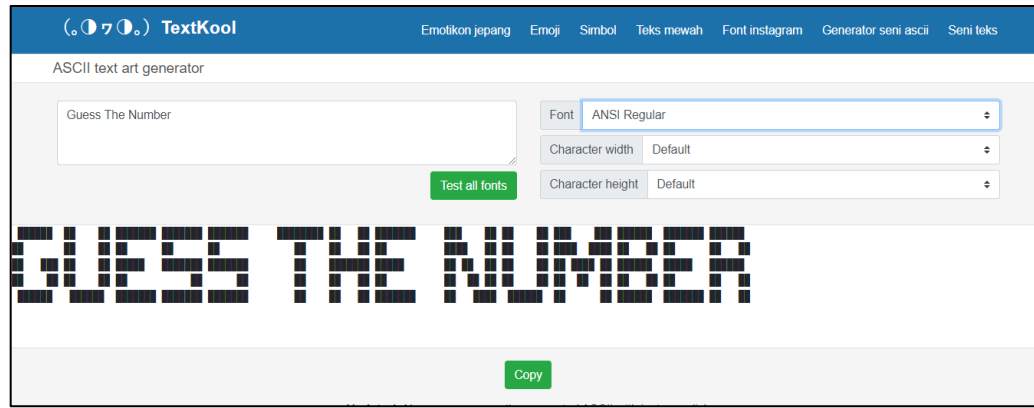
Tools

1. Pycharm

Class Structure

Duration	Topic
5 mins	Preparation
15 mins	Flow of the Project
10 mins	ASCII Text Art Generator
10 mins	Initial Setup
10 mins	Set Difficulty Function
10 mins	Check Lives Function
10 mins	Compare Function
15 mins	Game Function
5 mins	Wrap Up

Planned Activities	Resources
Preparation (5 mins)	
- Teacher greets students and ask how's their conditions. - Teacher instruct students to open their pycharm. - If possible, teacher can incorporate a mini game during this preparation time to engage with the students (optional).	
Flow of the Project (15 mins)	
- Teacher runs the program to show the students how the program works and start explaining the flow of the program to the students. - The first step of the program is to show the ASCII text art and then asking the user for the game difficulties. - After user chooses a difficulty, the program will show how many attempts the player has gotten depending on the chosen difficulties. - Therefore, the program will ask for the player's guess and it will keep on looping until the player gives the correct answer or the player ran out of attempts.	
ASCII Text Art Generator (10 mins)	
Show and Explain – ASCII Text Art Generator Teacher can start by showing the students examples of ASCII text art. Open the link provided in this section and start changing the fonts by using the generator to see different text arts.	Website: ASCII Text Art Generator



File/Code:

[Art.py](#)

- Teacher explain to the students that ASCII is a standard character encoding format for text data in computers and on the internet. Encoding is the process of putting a sequence of characters (letters, numbers, punctuation, and certain symbols) into a specialized format for efficient transmission or storage.
- Teacher explain to the students that in this case, we're using the ASCII encoding capability to generate text art for our project purpose.
- In this section, teacher can let the students choose their preferred text art for the project. But, teacher must notify the students to avoid text art that has backward slash “\” in it because it is an escape character in program. So, if we're using it the text art may not be shown as expected when we run our program.
- After choosing the preferred text art, teacher can instruct the students to create new folder on Pycharm for lesson 9 and create a python file in it. The file name can be something like **Art.py**
- Inside the file, create a variable to hold the text art that has been copied from the website. The variable can be named **logo**, and assign triple opening quotation mark to the variable. It will generate another triple quotation mark for the closing. Thus, teacher can order the students to paste the text art they selected to the variable.
- The location of the text art should be between the opening and closing triple quotation mark. For example: **logo = """ <insert text art here> """**

Initial Setup (10 mins)

Code and Explain – Initial Setup

- Teacher asks students to create a new python file in the same folder. This new python file will be the main project file. Students can name it `GuessTheNumber.py`
- After creating the project file, the first thing that we have to do in this section is importing modules. For this project we will be using random module for generating random numbers by using this code snippet: **import random**
- The next thing that we have to do is importing the text art that we created on the previous file. We can import the logo variable that holds the text art by using this code snippet: **from Art import logo**
- After importing the modules, the next thing that we have to do is initializing the variables that we will need for the project. Those are:
 - **random_number** : For storing random number generated by the random module by using **random.randint()**
 - **EASY** : For storing how many lives user can get if they're choosing easy level.
 - **MEDIUM** : For storing how many lives user can get if they're choosing medium level.
 - **HARD** : For storing how many lives user can get if they're choosing hard level.
- From this point, teacher can continue explaining about the functions that will be needed for the project and set that up. The functions that we will need:
 - **set_difficulty()** : This function will be responsible for setting the amount of attempts the player will get based on the player's input.
 - **check_lives()** : In this function, the program will check and show how many attempts the player has remaining and will return True if the students have no attempts remaining.
 - **compare()** : This function will be for comparing the player's guess with the correct answer, it will tell the player if the answer is correct, too high or too low.

File/Code:

[GuessTheNumber.py](#)

- **game()** : The game function will be the main function of the game, this function will contain the main game loop and function calls for the other functions.

Set Difficulty Function (10 mins)

Explain and Code – Set Difficulty Function

- This function will contain the input command that will get the player's input to choose the difficulties.
- The input has to be converted to lowercase to prevent errors when the user input it in capital or mixed.
- This function will also contain the conditional statement to assign the number of attempts depending on the player's input.
- Code together with the students for the input and difficulties conditional statements.

Check Lives Function (10 mins)

Explain and Code – Check Lives Function

- This function will contain the conditional statements to decide if the player has lives remaining or not.
- If the player still has lives remaining, the program will show how many lives the player has remaining.
- Let the students try to create the conditional statement based on the flow that already explained before.
- After they tried it, code together with the students for this.

Compare Function (10 mins)

Explain and Code – Compare Function

- This function's main job is to compare user's answer with the random number that is already generated and saved to the variable.
- There will be a simple logic in the form of conditional statements that returns print() statement with different messages according to three conditions:
 - If user's answer is greater than the actual random number that was generated, then this function will print "Too high" as a message.
 - If user's answer is lesser than the actual random number that was generated, then this function will print "Too low" as a message.
 - But if user's answer is the same with the actual random number that was generated, then this function will print "You are correct!" along with the number guessed.
- Code together with the students, this function will use comparison operators such as (==, < and >).

Game Function (15 mins)

Explain and Code – Game Function

- This function will be the main function of the game, it will contain the introduction text of the game, getting the amount of attempts using **set_difficulty()** function.
- The main loop of the function will use the while loop and the condition will be: as long as the player still has attempts remaining and the player has not answered correctly, the loop will keep going.
- Everytime it loops the attempts will be reduced by 1 and if the attempts reach 0 it will say you ran out of attempts.
- Let the students create the introduction text and create the main loop by themselves first.

- After they try it, code together with the students while explaining the function of each line of code.

Wrap Up (5 mins)

Wrapping Up

- Teacher sums up the class by asking questions, making sure the students remember the most important things.
- Teacher thanks the students for doing a great job overall.
- Teacher may also gives a hint on what will they do for the upcoming week.

☐ ☆ Good Luck With The Class !! ☆ ☐